# PyAnalySeries Reimagined: A Multi-Platform Revival of a Legacy Tool for Paleoclimate Time Series Analysis

Francisco Hevia-Cruz[1,2], Patrick Brockmann[1], Aline Govin[1], Elisabeth Michel[1]* and Didier Paillard[1]

[1]Laboratoire de Sciences du Climat et de l'Environnement LSCE-IPSL, CEA-CNRS-UVSQ, Université Paris-Saclay, Saint Aubin, France;
[2]Departamento de Ciencias de la Tierra, Facultad de Ciencias Químicas, Universidad de Concepción, Concepción, Chile

*elisabeth.michel@lsce.ipsl.fr

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## I. Introduction

The AnalySeries software [1] is well-known and widely distributed among the paleoclimate community. It was originally designed to analyze and correlate paleoclimatic time series, and has little competition due to its ease of use, user-friendly interface, and powerful analysis and processing tools. Nevertheless, due to support issues, it is quickly becoming obsolete.

We present here **PyAnalySeries**, a modern, **multi-platform** reimplementation of AnalySeries.

## II. Retention of key features of the original version and important enhancements

### Preserved features

- Interpolation-based correlation (previously "Linage")
- Signal filtering
- User-friendly visualization (Figs. 1-3)

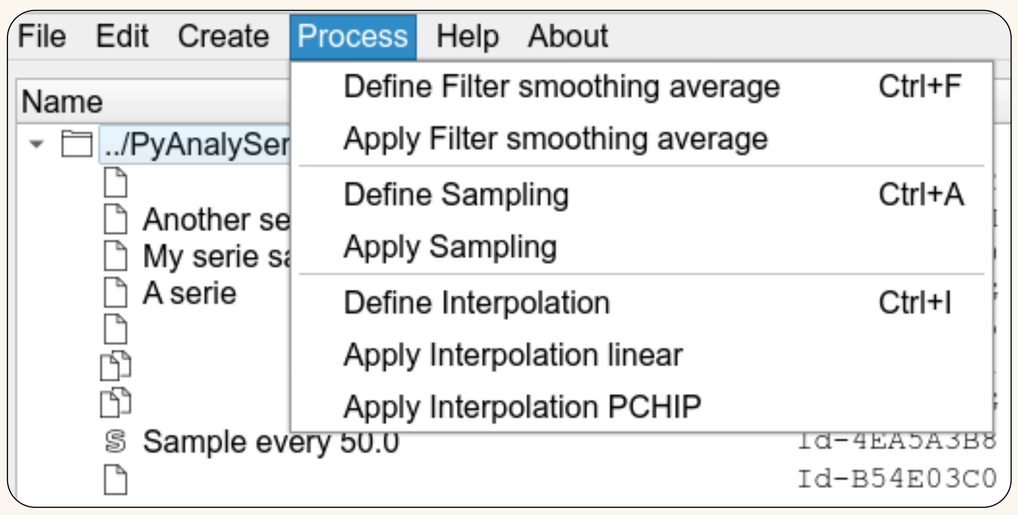



**Figure 1.** Screenshot of the main interface of **PyAnalySeries** with a **worksheet** opened with several series on it, and where the different menus can be seen. The **Process** menu is deployed, where a variety of processes are shown, including the **definition** and **application** of **Smoothing**, **Sampling**, and **Interpolation** functions.

### Improvements

- Written in Python (built on matplotlib, PyQt-based graphical interface) → Easily portable across platforms (Linux, macOS, and Windows)
- Redesigned to remain simple and intuitive, while adding interactivity (Figs. 2, 3) →
  - Dynamic visualization
  - Exportable figures
  - Keyboard shortcuts
  - Possibility to reopen and edit interpolation tie points
- Processing steps stored as .xlsx spreadsheet format files → Analyses are easily traceable, reproducible, and portable
- Additional functionalities →
  - Astronomical time series generation
  - Multi-panel plotting & saving (Fig. 3)
- "Define then Apply" workflow (Fig. 1) → Re-applying filters, samplings, or interpolations to new datasets

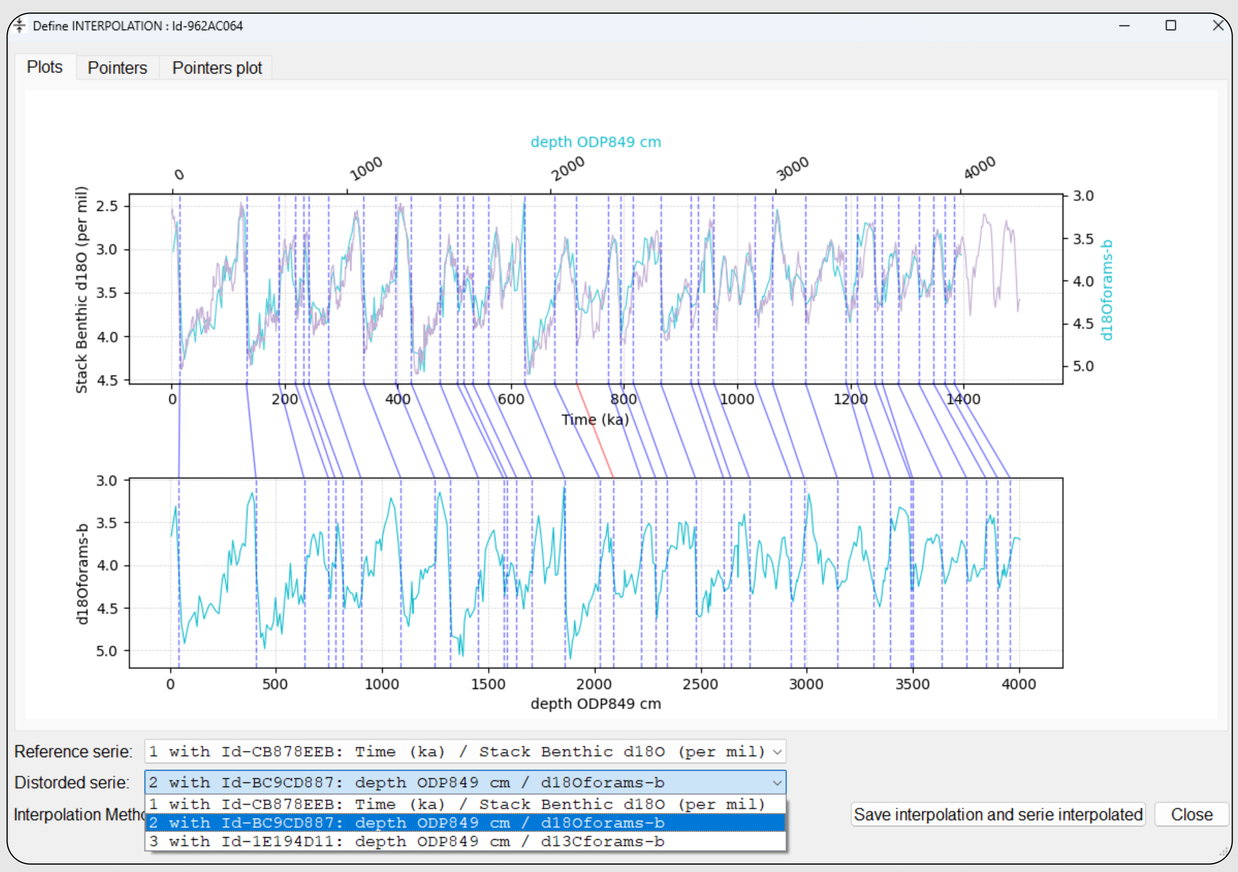



**Figure 2. Correlation** (Linage) from a reference time series over time (LR04 Stack; [2]) to a distorted series over depth ($\delta^{18}O$ and $\delta^{13}C$ from ODP849; [3]). The **simple graphics** of the original program are maintained. The use of keyboard **shortcuts** allows for **dynamic** correlations, while **improved graphics** allow for zooming in and out on both axes, observing changes in real time as **tie points** are added, and more. One of the main advantages of AnalySeries over other programs is the ability to compare different data for the same file ("**switch to**" function), which is preserved by PyAnalySeries.

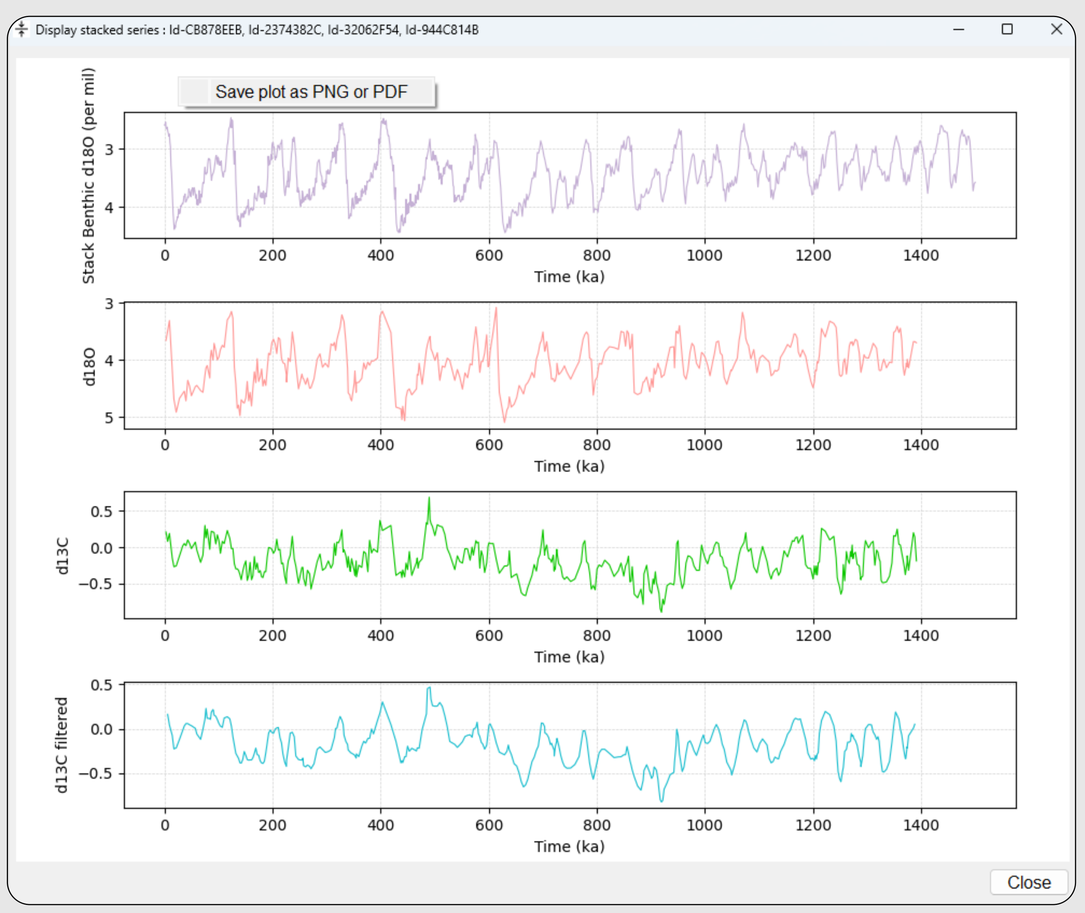



**Figure 3.** Example of the **enhanced graphics** capabilities of PyAnalySeries. This new version allows for generating **multiple graphics** (**Stacked**, as in the example, or **together** in a single graph with **single or multiple axes**). Graphs, including the correlation window (Fig. 2), can be **directly saved as PNG or PDF** by clicking on the right mouse button.

## III. Further development and perspectives

### Useful for any kind of 2D dataset

- PyAnalySeries can handle **any 2D dataset** requiring visual correlation and filtering → Potential extension to a **broader audience and other disciplines**

### Open Source, collaborative environment

- **Source code**, **documentation**, and **tutorials** are **freely available** at: **github**.com/PaleoIPSL/PyAnalySeries **zenodo**.org/records/15238083 → **Collaborative** & **evolving** development **environment**

YOU CAN ACCESS OUR **GIT-HUB REPOSITORY** DIRECTLY BY FLASHING THIS **QR** CODE!!

With **PyAnalySeries**, we aim to preserve the legacy of **AnalySeries** while adapting it to current **technological and scientific needs**

**References:** [1] Paillard et al. (1996). EOs. Trans. AGU, 77(39), 379-379. [2] Lisiecki and Raymo (2005). Paleoceanography, 20(1). [3] Mix et al. (1995). Earth Sci. Rev., 190, 190-223.

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